

Institute of Actuaries of India

Subject CB1-Business Finance

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u> A	[2 Marks]
<u>Solution 2:</u> D	[2 Marks]
<u>Solution 3:</u> A	[2 Marks]
<u>Solution 4:</u> A	[2 Marks]
<u>Solution 5:</u> D	[2 Marks]
<u>Solution 6:</u> D	[2 Marks]
<u>Solution 7:</u> A	[2 Marks]
<u>Solution 8:</u> D	[2 Marks]
<u>Solution 9:</u> C	[2 Marks]
<u>Solution 10:</u> B	[2 Marks]

Solution 11:

A stakeholder or stakeholder group for a company is any individual, entity or group of individuals or entities whose interests are affected by what the company does. [1]

Examples of Stakeholders

Lenders to a company are stakeholders because they have put money into the company, and wish to have the money repaid with interest. They will be concerned with anything that increases the risk of non-payment.

Employees of a company have an interest in what their company does. The company is their employer, and they are dependent on the company for their jobs, salaries, work experience, training, career prospects and so on. [1+1=2]

CSR is responsibility shown by a company for the broader interests of society as a whole, and for stakeholders other than shareholders. It includes concern for the environment, employees and society in general.

- It has been suggested that companies with CSR policies tend to be better managed and more successful commercially.
- From a business case perspective, improving CSR improves corporate reputation among stakeholders of the company. By improving reputation and stakeholder relations, the company is likely to perform better over the medium to long-term.
- Some institutional investors are required to report on the extent to which they take socially responsible investment into consideration when making investment decisions. There is a stronger probability of support from the investment community for a company with good CSR policies.
- Paying attention to CSR is important in terms of risk management. There are many risks related to poor CSR, including environmental and human rights related risks. Such risks can destroy reputation and impact on share value.
- CSR is also associated with conducting business in an ethical way. A company's business might benefit from the trust between suppliers and customers that comes from ethical dealing. [2]

[5 Marks]

Solution 12:

The first step is to identify the variables that should be considered in the simulation. [1]
 For example, interest rates and other economic variables may affect consumer spending and so those will have to be introduced as factors. Then the project's cash flows will have to be modelled using estimates based on the factors. [1]
 An attempt will have to be made to determine how an increase in interest rates might affect demand. This model will be complicated, otherwise there will be little point in using simulation. [1]
 The model should be run repeatedly until the average of the results obtained settle down. The model can be run with different assumptions to determine which assumptions are important in terms of the sensitivity of the results. [1]
 The results have to be analysed carefully because they will provide an indication of the distribution of expected returns. The wider the disparity between best, worst and most likely returns the greater the risk associated with this investment. [1]

[5 Marks]**Solution 13:**

In an LLP, the partners are liable to extent of their contributions to the LLP. In the case of a partnership firm, the partners are personally responsible to the creditors. By incorporating a LLP, the all the Partners can enjoy limited liability protection. [2]
 In case of a Partnership Firm, if the number of partners at any time reduces below the mandatory minimum of 2 due to death, incapacitation or resignation of a Partner, the partnership firm would stand dissolved. On the other hand, in case of a LLP, if the number of Partners reduces below 2, the sole Partner can still find a new Partner to fill the position without dissolution of the LLP. [2]

LLP is a lawful person and its subsistence does not depend on the partners. Hence, the partners of an LLP can keep changing from time to time, but it will not affect the existence, continuity or operations of the LLP. In a Partnership Firm, the death or resignation of a Partner would have serious consequences and the Partnership would have to be reconstituted. [1]

[5 Marks]**Solution 14:**

Investor perception is very important consideration for setting up dividends as this would have major impact on the share price of the company.

Further information required for setting dividend strategy during recession

- Whether the company can afford to pay the dividend
- When the profit of the company are going to recover to their former level
- Should the company pay scrip dividend in order to preserve cash
- How other companies in the sector are doing and the dividends they are paying.

Possible course of action

- Maintain the dividend despite fall in profits
- Declare Scrip dividend equal in size to the previous actual dividend
- Declare a postponement of dividend payment until the company is profitable again.
- Reduce the dividend pay-out

[5 Marks]**Solution 15:**

i) The Steps involved in the calculation and marks allotted for each of the steps are as follow.

1.Determination of right cash flow. If the candidate understands what a cash flow is or what should go into it. It should only have items that involve the movement of money. So, depreciation will not be considered. [4]

2. calculation of Discount factor

[1]

3. Calculation of Discounted Cash flow and NPV

[2]

Time	0	1	2	3	4	5
Revenue		2.5	2.5	2.5	2.5	2.5
Machine	-3					
Factory Deposit	-0.1					0.1
Raw Material	-0.2	-0.2	-0.2	-0.2	-0.2	
Inventory	-0.4	-0.4	-0.4	-0.4	-0.4	
Labour	-0.3	-0.3	-0.3	-0.3	-0.3	
Running Cost		-0.5	-0.5	-0.5	-0.5	-0.5
Dismantling						-0.3
Cash flow	-4	1.1	1.1	1.1	1.1	1.8
Discount Factor	1	0.935	0.873	0.816	0.763	0.713
DCF	-4.0	1.0	1.0	0.9	0.8	1.3
NPV	1.0					

[7]

- ii) The cash flows should be assessed in terms of the risk attached to the project. The risks borne by the company and its shareholders are not mirrored by those borne by the lender. [1.5]
 The risks to the company are based on the possibility that the project will not generate a satisfactory return in line with expectations, which has nothing to do with the risk that the bank will not be repaid on time or in full. [1.5]
 The bank will secure its advance against the company's other assets, which will increase risks to the existing providers of finance. [1.5]
 The loan will push up the cost of equity and so the real cost of the finance to the company will be higher than 7%. [1.5]

[6]

iii) Steps involved and marks

1. Running cost is increased correctly by 10% [1]
2. Discount factor is calculated correctly [2]
3. DCF and NPV calculated correctly. [2]

Time	0	1	2	3	4	5
Revenue		2.5	2.5	2.5	2.5	2.5
Machine	-3					
Factory Deposit	-0.1					0.1
Lease Payment	-0.2	-0.2	-0.2	-0.2	-0.2	
Inventory	-0.4	-0.4	-0.4	-0.4	-0.4	
Labour	-0.3	-0.3	-0.3	-0.3	-0.3	
Running Cost		-0.55	-0.55	-0.55	-0.55	-0.55
Dismantling						-0.3
Cash flow	-4	1.05	1.05	1.05	1.05	1.75
Discount Factor	1	0.917	0.842	0.772	0.708	0.650
DCF	-4.0	1.0	0.9	0.8	0.7	1.1
NPV	0.5					

[5]

- iv) Allowing for tax may affect the cost of capital because tax relief will be obtained on the loan. [2]

[20 Marks]

Solution 16:

- i) The prepares of the financial statements should avoid presenting an unduly optimistic set of results according to prudence concept.

The lowest reasonable figure should be stated for profit or for any assets and the highest reasonable figure should be state for any liabilities.

This means that there is very little danger of the figures lulling anybody into a false sense of security by overstating the company's strengths.

However, it is not permitted to include deliberate margins by understating assets or revenues or by overstating expenses or liabilities. [2]

- ii) Prudence should only be applied in situation where there is uncertainty.

A general insurance actuary set the reserves for outstanding claims liabilities for the business which is already written. These claims liabilities can be reported and/or unreported at the time of setting the reserves e.g. Outstanding reported claims and IBNR (Incurred but not reported claims).

Actuary should made provision for all known liabilities, expenses and losses ensuring there is no deliberate understatement of liabilities and there is no deliberate margin exists in the reserves amounts. [3]

[5 Marks]

Solution 17:

- Statistical approaches assume the continuation of established patterns and do not allow for changes in the environment being considered.
- Intuitive forecast use judgement and so can allow for future changes (although they would not necessarily do so).
- In particular, the experts involved in the forecast may also have some role in shaping future changes and can bring this to neat in their forecasts.
- Statistical approaches may be quicker and cheaper. Involving experts and getting them together or going through iterations of the Delphi technique is likely to be both costly and time-consuming.
- Statistical techniques may offer more rigor than judgement forecasts.
- In particular, they may use all relevant available past data, whereas intuitive forecasts may place more emphasis on recent events than longer-terms statistical trends.
- Statistical techniques may be less open to individual biases than judgement techniques although Delphi technique remove some of the biases that may exist in other forms of experts' opinions.

[5 Marks]

Solution 18:

- i) Gross profit is the difference between the selling prices of the goods and services which provide the basis for the company's main trading activities and the cost of sales.

Operating profit is defined as profit earned after all expenses except finance cost and interest.

Profit before tax is the operating profit adjusted for financing costs and income. Profit after tax will be calculated by deducting the tax from profit before tax. [3]

- ii) Gross profit gives an insight into the company's pricing policies.

If the company is incurring and experiencing high distribution costs and administrative expense can have gross profit but no operating profit e.g. advertising expense, employees' salaries [2]

[5 Marks]

Solution 19:

For analysis any business efficiency, I will analysis below three ratios -

Inventory turnover period = $\text{Inventories} \times 365 / \text{Cost of sales}$

This ratio indicates to show how long inventory is held for an average. An inventory turnover period that is less rapid than other companies in the same industry might indicate an inefficiently large inventory holding. [2]

Trade receivable turnover period = $\text{Trade receivables} \times 365 / \text{Credit sales}$

This ratio measures average length of time taken for trade receivable to settle their balance. It is desirable for this period to be as short as possible as it will be better for the company's cash-flow. [2]

Trade payable turnover period = $\text{Trade payables} \times 365 / \text{Credit Purchases}$.

This ratio indicates the average number of credit that a company has from its suppliers. A high ratio may indicate that the company is able to obtain a long credit period from its suppliers, which will be of benefits to its cash-flows. [2]

[6 Marks]

Solution 20:

- i) ABC Ltd. Income statement for the year ended 31st March 2019

Revenue	1,500,000
Cost of Sales	(647,000)
Gross Profit	853,000
Administrative Staff wages	(3,000)
Distribution Cost	(266,000)
	584,000
Interest	(10,000)
Profit for the year	574,000

[9]

- ii) ABC Ltd. Statement of changes in equity for the year ended 31st March 2019

	Share Capital	Revaluation Reserve	Retained Earnings	Total
Opening Balance	200,000	180,000	250,000	630,000
Revaluation		540,000		540,000
Profit for the year			574,000	574,000
Dividend Paid			(45,000)	(45,000)
Closing Balance	200,000	720,000	779,000	1,699,00

[4]

- iii) ABC Ltd. Statement of financial position as at 31st March 2019

Non-current Assets	
Property, Plant and Equipment	1,672,000
Current Assets	
Inventory	30,000
Trade Receivables	210,000
	240,000
Total Assets	1,912,000
Equity and Liabilities	
Equity	
Share Capital	200,000
Revaluation Reserve	720,000
Retained Earnings	779,000
	1,699,000
Non-current Liability	
Loan	145,000
Current Liability	
Trade Payables	50,000
Bank	18,000
	68,000
Total Equity and Liabilities	1,912,000

Notes:

1) Property, Plant and Equipment

Cost or Valuation	Land	Building	Machinery	Vehicles	Total
Opening Balance	600,000	400,000	150,000	325,000	1,475,000
Revaluation	200,000	300,000			500,000
Closing Balance	800,000	700,000	150,000	325,000	1,975,000

2) Depreciation

Cost or Valuation	Land	Building	Machinery	Vehicles	Total
Opening Balance		40,000	80,000	170,000	290,000
Revaluation		(40,000)			(40,000)
Closing Balance		7,000	15,000	31,000	53,000
		7,000	95,000	201,000	303,000
Net book value	800,000	693,000	55,000	124,000	1,672,000

3) Cost of Sales

Cost of Inventory Consumed	350,000
Factory running costs	100,000
Manufacturing wages	175,000
Depreciation of buildings	7,000
Depreciation of Machinery	15,000
	647,000

4) Cost of Distribution

Advertising Cost	55,000
Delivery Vehicle running costs	60,000
Sales Salaries	120,000
Depreciation of Vehicles	31,000
	266,000

[19 Marks]
